ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu [ANSWER KEY]

Unit 6.01: Vector Fields, Conservative Fields & Line Integrals

Q1. Consider the vector field mathbf{F}x,y = langle 3x, 2y rangle . Evaluate the vector field at the point 2, 4 .

[x] A: langle 6, 8 rangle

[ ] B: langle 8, 6 rangle

[ ] C: langle 5, 6 rangle

[ ] D: langle 6, 5 rangle

Solution: The vector field is given by mathbf{F}x,y = langle 3x, 2y rangle . To evaluate the vector field at the point 2, 4 , we substitute x=2 and y=4 into the components of the vector field. First component: 3x = 32 = 6 . Second component: 2y = 24 = 8 . Therefore, mathbf{F}2,4 = langle 6, 8 rangle .

Q2. Consider the vector field mathbf{F}x, y = langle 2x, 3y rangle . Which of the following options best represents the vector field at the point 1, 1?

[x] A: langle 2, 3 rangle

[ ] B: langle 1, 1 rangle

[ ] C: langle 4, 6 rangle

[ ] D: langle 2, 6 rangle

Solution: To find the representation of the vector field mathbf{F}x, y = langle 2x, 3y rangle at a specific point, we substitute the coordinates of the point into the components of the vector field. At the point 1, 1, we have: x = 1 and y = 1 . Substituting these values into mathbf{F}x, y gives: mathbf{F}1, 1 = langle 21, 31 rangle = langle 2, 3 rangle . Therefore, the vector field at the point 1, 1 is langle 2, 3 rangle . Option B is incorrect as it represents the position vector at 1,1. Option C is incorrect as it represents the vector field at 2,2. Option D is incorrect as it uses the x-component from 2,y and the y-component from x,2.

Q3. Consider the vector field mathbf{F}x,y = langle 2x+y, x-3y rangle . What is the value of the vector field at the point 3, -1 ?

[ ] A: langle 5, 6 rangle

[ ] B: langle 5, 0 rangle

[ ] C: langle 7, 6 rangle

[x] D: langle 7, 0 rangle

Solution: To find the value of the vector field mathbf{F}x,y = langle 2x+y, x-3y rangle at the point 3, -1 , we substitute x=3 and y=-1 into the components of the vector field. The first component is 2x+y = 23 + -1 = 6 - 1 = 5 . The second component is x-3y = 3 - 3-1 = 3 + 3 = 6 . Therefore, the vector field at 3, -1 is langle 5, 6 rangle .

Q4. Evaluate int_C x^{2},ds where C is the segment from left 0, 0right to left 5, 12right .

[ ] A: frac{25}{3}

[ ] B: frac{4225}{3}

[ ] C: frac{328}{3}

[x] D: frac{325}{3}

Solution: mathbf rt=langle 5 t, 12 t rangle, ds=13,dt . intlimits_{0}^{1} 325 t^{2}, dt = frac{325}{3} .

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Q5. Which vector field is conservative on all of mathbb R^2 ?

[x] A: langle e^{y}, x e^{y} rangle

[ ] B: langle x y, x rangle

[ ] C: langle x + y, - x + y rangle

[ ] D: langle x^{2} y, x y^{2} rangle

Solution: On mathbb R^2 , langle P,Qrangle is conservative iff P_y=Q_x . For langle e^{y}, x e^{y} rangle : P_y=e^{y}=Q_x ; each other field has P_yne Q_x .

Q6. Evaluate int_Cmathbf Fcdot dmathbf r for mathbf F=langle y^{2}, 2 x rangle along the segment from left -2, -2right to left -2, 3right .

[ ] A: 20

[ ] B: -19

[ ] C: -40

[x] D: -20

Solution: Parametrize mathbf rt=langle -2, 5 t - 2 rangle, 0le tle1 . Then int_0^1mathbf Fmathbf rtcdotmathbf r't,dt=intlimits_{0}^{1} left-20right, dt=-20 .

Q7. Evaluate int_C x + y,ds where C is the segment from left 0, 0right to left 4, 3right .

[x] A: frac{35}{2}

[ ] B: frac{7}{2}

[ ] C: frac{175}{2}

[ ] D: frac{37}{2}

Solution: mathbf rt=langle 4 t, 3 t rangle, ds=5,dt . intlimits_{0}^{1} 35 t, dt = frac{35}{2} .

Q8. Evaluate int_Cmathbf Fcdot dmathbf r for mathbf F=langle x y, y rangle along the segment from left 1, 1right to left 1, -2right .

[ ] A: - frac{3}{2}

[ ] B: frac{5}{2}

[x] C: frac{3}{2}

[ ] D: 3

Solution: Parametrize mathbf rt=langle 1, 1 - 3 t rangle, 0le tle1 . Then int_0^1mathbf Fmathbf rtcdotmathbf r't,dt=intlimits_{0}^{1} left9 t - 3right, dt=frac{3}{2} .